

1) IP Address of the Device

- IPv4 Address: 192.168.1.10
- IP Version: IPv4
- IP Type: Private
- A private IP address is used for communication inside a local network and is normally not directly reachable from the public Internet.



```
Command Prompt

ipconfig

IPv4 Address . . . . . : 192.168.1.10
Address Type           : IPv4
IP Type                 : Private
```

2) Valid Private and Public IPv4 Addresses

- 192.168.1.10 — Valid Private IP
- 10.0.0.5 — Valid Private IP
- 172.16.5.20 — Valid Private IP
- 8.8.8.8 — Valid Public IP
- 123.45.67.89 — Valid Public IP

IP Address	Type
192.168.1.10	Private
10.0.0.5	Private
172.16.5.20	Private
8.8.8.8	Public
123.45.67.89	Public

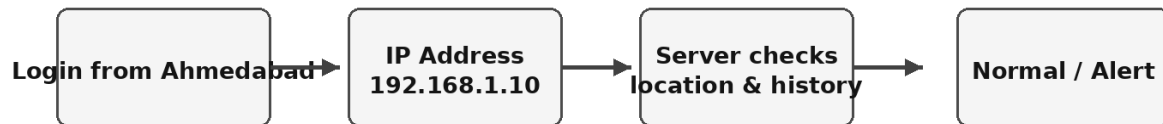
3) IPv4 Validation in Python

- The script accepts an IP address as input.
- It checks that the address contains exactly four parts.
- Each part must contain only digits from 0 to 255.
- No third-party library is used.

```
def valid_ipv4(ip):  
    parts = ip.split('.')  
    if len(parts) != 4: return False  
    for p in parts:  
        if not p.isdigit() or not 0 <= int(p) <= 255: return False  
    return True  
  
print(valid_ipv4(input('IP: ')))
```

4) IP Addresses and Suspicious Login Detection

- A service can record the IP address used during a login.
- The IP can be mapped approximately to a network or geographic location.
- A login from a new or unusual location can be compared with previous login activity.
- If the change is unusual, the service can request verification or alert the user.



IP address helps estimate the login network/location